Regis University
College for Professional Studies
School of Computer & Information Sciences

SYLLABUS

COURSE NUMBER: CS 390

COURSE TITLE: Principles of Programming Language

Course Description

CS 390. Principles of Programming Language (3). Introduces the constructs upon which contemporary programming languages are based. Students investigate programs written in declarative and imperative programming languages including functional, logic, structured, and object-based approaches.

Course Prerequisites

CS372 Data Structures & CS375 Computation Theory

Course Outcomes

Successful completion of this course will enable student to do the following:

1. Describe distinguishing characteristics of declarative (functional & logical) and imperative (procedural & object-oriented) programming language paradigms and explain how these characteristics manifest in historic and contemporary programming languages.
2. Evaluate syntactic, semantic, and pragmatic tradeoffs among the various programming paradigms and different programming languages.
3. Summarize the history and continuing evolution of programming languages.
4. Demonstrate different forms of declaration, typing, binding, visibility, scoping, and lifetime management for various programming language constructs (e.g variables, functions, etc.).
5. Use formal systems, including Lambda Calculus, to explain and model various programming language concepts.
6. Develop and analyze programs written in the various programming paradigms.
7. Choose an appropriate programming language solution for a given programming task.

Course Materials and Resources, including Technology Tools

Required:

Sebesta, R. W. (2010). *Concepts of programming languages (9th ed.).* Boston, MA: Addison-Wesley (Pearson Education). ISBN13: 978-0-13-607347-5 or ISBN10: 0-13-607347-6 hardcover.

Tate, B. A. (2010). *Seven languages in seven weeks: A pragmatic guide to learning programming languages (1st ed.).* Raleigh, NC: Pragmatic Bookshelf (O'Reilly Media). ISBN13: 978-1-934356-59-3 or ISBN10: 1-934356-59-X softcover.

Required Supplemental Material:

Supplemental Weekly Readings (given out in class and online).

Required Software:

SWI-Prolog
(free at: http://www.swi-prolog.org)
Haskell
(free at: http://www.haskell.org/haskellwiki/Haskell)
Ruby
(free at: http://www.ruby-lang.org/en/downloads/)

Method of Instruction

This course presents material using classroom lectures or online readings, textbook readings, facilitated group discussions, and peer feedback. Students are assessed via individual assignments, a group assignment, exams, and discussion participation.

Related Standards

Related Standards	
Program Outcomes	
	This course enables students to achieve the following Program outcomes by the time of graduation:
	An ability to analyze a problem, and identify and define the computing requirements appropriate to its solution
	An ability to design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs
	Recognition of the need for and an ability to engage in continuing professional development
	An ability to communicate effectively with a range of audiences
	An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices
Regis Nine Educational Outcomes	
	Content Knowledge in Discipline
	Diverse Cultures, Perspectives, and Belief Systems
	Arts, Sciences, and Humanities
	Critical Thinking
	Communication
	Technology
	Ethical and Social Responsibility
	Leadership in Service to Others
	Lifelong Learning
Jesuit Themes - These are themes that are embedded as part of our interactions and instruction at Regis, but are not written in as activities	
	Magis, Finding God in All Things, Unity of Hearts and Mind, Contemplatives

	in Action, Care of the Whole Person, Men and Women for Others

Initial First Week Assignment

- Read Chapter 1 of the Sebesta and Tate textbooks (see below).

Course Overview Grid

Week	Learning Topics	Reading Assignments*	Assignments Due*
1	LT 1: Overview & History LT 4: Language Design Paradigms, Imperative LT 6: Type Systems LT 8: Languages in Practice: Ruby	Sebesta, R. W., (2010): Chpt. 1 and §6.1, §6.6, §6.10 – §6.12 Tate, B.A., (2010): §1.0 – §1.1, §2.0 – §2.3 Handouts: (i) Introduction, (ii) Motivational design examples, (iii) Paradigms, (iv) Motivation type examples, (v) Types Systems: Part I (vi) Programming Guidelines	Assignment #1 **Due**: *Classroom*: Start of Wk 2 class *Online*: End of Wk 1
2	LT 2: Syntax LT 4: Language Design: Object-Oriented LT 7: Lambda Calculus Syntax LT 8: Languages in Practice: Ruby	Sebesta: §3.1-:§3.4, §4.1-:§4.5 §12.1-:§12.2 & §12.9 Tate: §2.3 – §2.5. Handouts: (i) Syntax, (ii) Attribute Grammar, (iii) Lambda Calculus I	Assignment #2 **Due**: *Classroom*: Start of Wk 3 class *Online*: End of Wk 2
3	LT 3: Semantics Overview & Operational LT 4: Language Design Declarative & Logical LT 7: Lambda Calculus Semantics LT 8: Languages in Practice: Prolog	Sebesta: Chapter 16 Tate: §4.1 – §4.2 Handouts: (i) Semantics, (ii) Denotational Semantics (iii) Lambda Part II, III, IV	Assignment #3 **Due**: *Classroom*: Start of Wk 4 class *Online*: End of Wk 3
4	LT 5: Declarations LT 4: Language Design Scripting	Sebesta: Chapter 5 & §2.18 Handouts (i) Declarations (ii) Scripting Languages	Midterm Exam **Due**: *Classroom*: Take Home *Online*: End of Wk 4
5	LT 3: Semantics Denotational	Sebesta: §3.5	Assignment #4 **Due**:

	LT 8: Languages in Practice: Prolog	Tate: §4.3 – §4.5 Handouts: (i) Denotational Semantics (ii) Semantic Algebra (iii) Examples	*Classroom*: Start of Wk 6 class *Online*: End of Wk 5
6	LT 4: Language Design Functional LT 7: Lambda Calculus Encoding & Applied LT 8: Languages in Practice: Haskell	Sebesta: Chpt. 15 Tate: §8.1-:§8.2 Handouts: (i) Funtional Programming (ii) Lambda Encoding (iii) Applied Lambda Calculus	Assignment #5 **Due**: *Classroom*: Start of Wk 7 class *Online*: End of Wk 6
7	LT 6: Type Systems Formal System LT 7: Lambda Calculus Typed Calculus LT 8: Languages in Practice: Haskell	Sebesta: §6.10-:§6.12 Chapter 11 Tate: §8.3-:§8.5 Handout: (i) Typed Lambda Calculus	Assignment #6 **Due**: *Classroom*: Start of Wk 8 class *Online*: End of Wk 7
8	LT 1: Overview & History LT 4: Language Design** design criteria abstract data type object-oriented, concurrency, exceptions	Reading** Sebesta: §1.6 Sebesta: §11.1 - §11.4 Sebesta: §12.1 - §12.2, §12.9 Sebesta: §13.1- §13.2 Sebesta: §14.1, §14.4 - §14.6 Handouts: (i) Design Criteria	Final Exam **Due**: *Classroom*: In class *Online*: End of Wk 8

****Note to Classroom sections only:*** Exact dates for reading assignments and homework assignments will be included in your facilitator's syllabus, handed out the first night of class.

** These language design topics are also introduced and discussed in-depth in other courses.

Student Evaluation Grid

Assignment	Value (percent of overall course grade)
Weekly Course Participation	10%
Programming Assignments (6)	60%
Midterm	15%
Final exam	15%
TOTAL	**100 %**

Regis University Grading Scale

A	93 – 100	C	73 - 76
A-	90 – 92	C-	70 – 72
B+	87 – 89	D+	67 – 69
B	83 – 86	D	63 – 66
B-	80 – 82	D-	60 – 62
C+	77 – 79	F	59 or Lower

Exams

There will be a midterm exam and a final exam. Exams may be timed; they might be open book, open notes, closed book, and/or closed notes. They might be in-class or take home. Exam questions will be cumulative, taken from reading assignments and class presentations and focus primarily on principles and concepts, as opposed to weekly programming assignments.

Attendance/Participation

Facilitator will provide specific attendance/participation policies in accordance with the official attendance policy of the School of Computer and Information Sciences.

Attendance Policy of the School of Computer and Information Sciences

Student participation is essential. Students are responsible for all material covered in each week. Students' interaction is central to the learning experience and imparts critical observations and knowledge to all concerned.

If a student misses more than one week of class participation without prior faculty approval, the facilitator may assign a failing grade.

Late Assignments/Assignment Revisions

Facilitator will provide specific late assignments/assignment revisions policies not to exceed the maximum official policy of the School of Computer and Information Sciences.

Late Assignments/Assignment Revisions Policy of the School of Computer and Information Sciences

Occasionally, work, health, and family obligations have priority over course work. If you experience these issues, please notify the facilitator in advance. If an assignment is due, please make every effort to submit your assignment on time. In the case of a negotiated absence or delay, assignments turned in by the agreed upon due date will receive full credit.

If you have not negotiated with the facilitator, assignments turned in late will be graded, then reduced by 3% per day. Assignments **will not** be accepted later than one week after due date. Week 8 assignment may only be 3 days late.

Incomplete Grade
An incomplete grade is given at the discretion of the facilitator. A facilitator is not required to assign an incomplete grade. Each occurrence will be negotiated on an individual basis.

Incomplete Grade Policy of the College of Professional Studies
A grade of Incomplete or 'I" denotes that the required work for the course is incomplete due to unforeseen circumstances. Unforeseen circumstances mean, for example, that an accident, an illness, a death, or a major life transition has occurred. This grade is awarded at the discretion of the instructor and is submitted with an alternate grade of "I/F."

A student must submit a written request to the instructor asking that an incomplete grade be assigned prior to the end of the term in which the course is taken. The course instructor will determine the length of time for course completion, and the remaining requirements to complete the course; however the maximum time for completion is the end of the following semester.

Faculty will notify students in writing of the approval/denial of their request and provide instructions and guidelines required to resolve the incomplete grade. If the course work is not completed by the end of the next applicable semester and a grade change form submitted by the instructor, the incomplete grade reverts to the alternate grade assigned by the instructor and is calculated in the grade point average.

Disability Statement:
Equal access to education means equal opportunity to learn in an academic setting. Under Section 504 of the 1973 Rehabilitation Act and the Americans with Disabilities Act of 1990, students with documented disabilities who are otherwise qualified have a right to equal access to classes at Regis University. If you have a documented disability requiring academic adjustments for this class, please contact the Director of the Office of Disability Services at **303-458-4941** for discussion of disability issues. Since accommodations are not retroactive, it is suggested that you contact the Disability Services Office as soon as possible; adequate lead-time may be required to receive accommodations, including review of appropriate documentation.

Academic Dishonesty
Regis University is committed to intellectual integrity in its academic pursuits. Sanctions may, therefore, be imposed by the College for Professional Studies faculty or schools for cheating (defined as using inappropriate sources of information on a test) or plagiarism (defined as presenting as one's own, the ideas, words, or products of another). Such sanctions may include a failing grade on the assignment, failure of the course, or expulsion of the student from the course or the department.

Non-Public, Confidential or Proprietary Information Compliance
For class projects that relate to a student's employment, the student is responsible for complying with all applicable policies of the employer, including those related to non-public, confidential or proprietary information